

A Cruel Lesson in Probability: How Uncertainty Definitely Got Me Fired

A refresher on uncertainty, sampling, and inference

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The Night Shift

My first job: the night shift at a dice factory

- Like most 14-year-olds, I had just graduated from eighth grade and working the night shift at a die factory.
- I was in charge of monitoring a machine that produced six-sided fair dice.
- One school night, I dozed off during my shift but woke when the machine made a loud *thud*.
- The next batch looked normal, just to be sure I rolled one die a bunch (30 times) and declared that there was “no difference.”

Were 30 rolls enough to detect a small bias?

Set

A *set* is a collection of distinct objects, called *elements*. Sets are usually denoted by capital letters and their elements by lowercase letters.

Examples

- $X = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$, so $\frac{1}{2} \in X$.
- $Z = \{0, 1, 2, 3, \dots\}$ is the set of nonnegative integers.
- $C = \{\text{Picard, Janeway, Sisko, Kirk, } \dots, \text{Pike}\}$.

Set Theory 2

Finite and countably infinite sets

- A set is **finite** if it has n elements for some nonnegative integer n .
- A set is **countably infinite** if its elements can be put in one-to-one correspondence with $\mathbb{N} = \{1, 2, 3, \dots\}$.
- A set is **countable** if it is finite or countably infinite.

Examples

- $C = \{5, 6, 7, 8, 9, 10, 11, 12\}$ is finite and $|C| = 8$.
- $\mathbb{N}$ is countably infinite.
- $\mathbb{R}$ is uncountable.

Set Theory 3

Experiments and outcomes as sets

- A **random experiment** is repeatable under similar conditions, but its individual outcome is uncertain.
- The set of all possible outcomes is the **sample space**, conventionally denoted Ω .
- Coin toss: $\Omega = \{H, T\}$.
- Six-sided die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- An **event** is a subset of the sample space.

Set Theory 4

Rolling two dice

We roll a red die and a white die. The outcome is an ordered pair,

$$\Omega = \{(r, w) : r, w \in \{1, 2, 3, 4, 5, 6\}\}, \quad |\Omega| = 36.$$



Set Theory 5

Rolling two dice: R practice

- Simulate 100 rolls of a red die and a white die.
- Store each trial as an ordered pair (r, w) .
- Plot the marginal counts for each die.

Can we code this in R?

Set Theory 5.1

Rolling two dice: R answer

```
set.seed(2025)
n <- 100

rolls <- data.frame(
  red   = sample(1:6, n, replace = TRUE),
  white = sample(1:6, n, replace = TRUE)
)

head(rolls)                # ordered outcomes
table(rolls$red)
table(rolls$white)

par(mfrow = c(1, 2))
barplot(table(rolls$red), col = "red")
barplot(table(rolls$white), col = "white")
```


Set Theory 6

Set notation

- The empty set is $\emptyset$.
- The complement of $A \subseteq \Omega$ is $A^c = \{\omega \in \Omega : \omega \notin A\}$.
- $A \subseteq B$ means every element of A is also in B .
- $A \cup B$ contains outcomes in A or B (or both).
- $A \cap B$ contains outcomes in both A and B .
- A and B are **disjoint** if $A \cap B = \emptyset$.

Set Theory 7

A 20-sided die example

Let $\Omega = \{1, 2, \dots, 20\}$ and define

$$\begin{aligned}W &= \{19\}, & X &= \{1, 5, 10\}, \\Y &= \{5, 12, 14, 20\}, & Z &= \{8, 12, 20\}.\end{aligned}$$

Set operations

$$\begin{aligned}X^c &= \{2, 3, 4, 6, 7, 8, 9, 11, 12, \dots, 20\}, \\X \cup Y &= \{1, 5, 10, 12, 14, 20\}, \\X \cap Y &= \{5\}, & W \cap Z &= \emptyset, \\(X \cup Y) \cap (X \cup Z) &= X \cup (Y \cap Z) \\&= \{1, 5, 10, 12, 20\}.\end{aligned}$$

Function

A function $f : X \rightarrow Y$ assigns each input $x \in X$ exactly one output $f(x) \in Y$.

- X is the **domain**.
- Y is the **codomain**.
- The set $\{f(x) : x \in X\}$ is the **range** of f .

$$x \mapsto f(x) = y, \quad f : X \longrightarrow Y.$$

Probability 1

A mathematical model of uncertainty

A probability space is a triple $(\Omega, \mathcal{F}, P)$, where Ω is the sample space, $\mathcal{F}$ is a collection of events, and

$$P : \mathcal{F} \longrightarrow [0, 1]$$

assigns a probability to each event.

Kolmogorov axioms

For events $A, A_1, A_2, \dots \in \mathcal{F}$:

- 1 $P(A) \geq 0$.
- 2 $P(\Omega) = 1$.
- 3 If $A_i \cap A_j = \emptyset$ whenever $i \neq j$, then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

If an experiment is repeated n times, the relative frequency of event A is

$$\hat{P}_n(A) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\omega_i \in A\}.$$

Under repeated independent trials, $\hat{P}_n(A)$ approaches $P(A)$ as n grows.

```
set.seed(2025)
n <- 1000
red <- sample(1:6, n, replace = TRUE)

# A = "the red die is even"
running_frequency <- cumsum(red %% 2 == 0) / seq_len(n)

plot(running_frequency, type = "l")
abline(h = 1/2, col = "red", lty = 2)
```

For events $A, B \in \mathcal{F}$:

Additional properties

$$P(A^c) = 1 - P(A),$$

$$P(\emptyset) = 0,$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are disjoint, the last identity reduces to

$$P(A \cup B) = P(A) + P(B).$$

Probability 4

Example with a fair die

- Sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Event space: $\mathcal{F} = 2^\Omega$, the set of all subsets of Ω .
- For a fair die, $P(\{i\}) = 1/6$ for each i .

Let $A = \{2\}$ and $B = \{4\}$. Since $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

The singleton events $\{1\}, \dots, \{6\}$ are pairwise disjoint and exhaustive because their union is Ω .

Definition

For $P(B) > 0$, the conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Return to the fair die. Let $E = \{2, 4, 6\}$, $A = \{2\}$, and $B = \{4\}$. Given that the roll is even,

$$P(A \mid E) = \frac{1}{3}, \quad P(B \mid E) = \frac{1}{3}, \quad P(A \cup B \mid E) = \frac{2}{3}.$$

Political science translation

$P(\text{turnout} \mid \text{campaign contact})$ asks about turnout among voters who were contacted, not among voters overall.

Independent events

Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Equivalently, when the relevant probabilities are positive,
 $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$.

For a fair die, let $A = \{1, 2, 3\}$ and $B = \{3, 4\}$. Then

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{3}, \quad P(A \cap B) = P(\{3\}) = \frac{1}{6} = P(A)P(B).$$

Thus A and B are independent, even though they are not disjoint.

Political science translation: Party identification and turnout are independent only if learning a person's party identification does not change the probability assigned to turnout.

Random variable

A random variable is a measurable function

$$X : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

- It assigns a real number to each outcome in the sample space.
- Die roll: $X(\omega)$ is the number on the upper face.
- Coin toss: we might code $H = 1, T = 0$ (or $H = 1, T = -1$).

Probability mass function

A discrete random variable takes values in a finite or countably infinite set D . Its probability mass function (pmf) is

$$p_X(x) = P(X = x), \quad x \in D.$$

Every pmf satisfies

$$p_X(x) \geq 0, \quad \sum_{x \in D} p_X(x) = 1.$$

The support is $\text{supp}(X) = \{x : p_X(x) > 0\}$.

Probability 9

A fair die as a random variable

Let X be the upper face of a fair six-sided die.

$$\text{supp}(X) = \{1, 2, 3, 4, 5, 6\}, \quad p_X(i) = \frac{1}{6}.$$

The cumulative distribution function (CDF) is $F_X(x) = P(X \leq x)$.
For example,

$$F_X(0) = 0, \quad F_X(1.5) = \frac{1}{6}, \quad F_X(2) = \frac{1}{3}, \quad F_X(6) = 1.$$

Values outside the support, such as 0, have probability mass zero.

Probability 10

The pmf of a fair die

$$p_X(x) = P(X = x) = \begin{cases} \frac{1}{6}, & x \in \{1, 2, 3, 4, 5, 6\}, \\ 0, & \text{otherwise.} \end{cases}$$

- The support is $\text{supp}(X) = \{1, 2, 3, 4, 5, 6\}$.
- The pmf records probability at points; the CDF accumulates probability up to a point.

For one fair coin toss, let $X = 1$ for heads and $X = 0$ for tails.

$$p_X(x) = P(X = x) = \begin{cases} \frac{1}{2}, & x = 0, \\ \frac{1}{2}, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hence $X \sim \text{Bernoulli}(1/2)$ and

$$\text{supp}(X) = \{0, 1\}.$$

Political science translation

Turnout, vote choice, and whether a respondent supports a proposal can each be represented as Bernoulli variables when coded 1 for the event and 0 otherwise.

Probability 12

Expected value is a probability-weighted average

Expected value

For a discrete random variable,

$$\mu = \mathbb{E}[X] = \sum_{x \in \text{supp}(X)} x p_X(x).$$

For a continuous random variable with density f_X ,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx,$$

provided the expectation exists.

- Expected value is the long-run average over repeated realizations.
- It need not be a value that the variable can actually take.

Probability 13

Expected outcomes in familiar examples

Fair die

$$\mathbb{E}[X] = \sum_{x=1}^6 x \left(\frac{1}{6}\right) = \frac{21}{6} = 3.5.$$

The expected value is 3.5, although a die can never show 3.5.

Linearity gives $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$ and $\mathbb{E}[\sum_i X_i] = \sum_i \mathbb{E}[X_i]$; independence is not required.

Turnout indicator

Let $T = 1$ if a voter turns out and $T = 0$ otherwise. If $P(T = 1) = p$, then

$$\mathbb{E}[T] = 0(1 - p) + 1(p) = p.$$

For n voters, the expected number who turn out is np .

Probability 14

Variance measures spread around the expectation

Variance and standard deviation

If $\mu = \mathbb{E}[X]$, then

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

The standard deviation is

$$\sigma = \sqrt{\text{Var}(X)}.$$

- Variance is nonnegative and is measured in squared units.
- Standard deviation is in the same units as X .
- Adding a constant does not change variance; rescaling does:
 $\text{Var}(a + bX) = b^2 \text{Var}(X).$

Probability 15

Averages become more stable as the sample grows

For a turnout indicator $T \sim \text{Bernoulli}(p)$,

$$\mathbb{E}[T] = p, \quad \text{Var}(T) = p(1 - p).$$

If $X_1, \dots, X_n$ are independent with common mean μ and variance σ^2 , then

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

For a sample proportion $\hat{p} = \bar{T}$,

$$\mathbb{E}[\hat{p}] = p, \quad \text{Var}(\hat{p}) = \frac{p(1 - p)}{n}.$$

Probability 16

From Bernoulli trials to a binomial count

Toss a fair coin independently ten times. Let X be the number of heads.

- Number of independent trials: $n = 10$.
- Success probability on each trial: $p = 1/2$.
- Desired number of successes: $k = 8$.

What is $P(X = 8)$?

Probability 17

Permutations and combinations

- An ordered selection of k objects from n distinct objects is a permutation:

$${}_nP_k = \frac{n!}{(n-k)!}.$$

- An unordered selection of k objects from n distinct objects is a combination:

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}.$$

The binomial coefficient $\binom{n}{k}$ counts which k of the n trials are successes.

If X counts successes in n independent Bernoulli trials with success probability p , then

$$X \sim \text{Binomial}(n, p)$$

and

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

The formula combines the probability of one success/failure pattern with the number of patterns containing exactly k successes.

Probability 19

Eight heads in ten fair tosses

For $X \sim \text{Binomial}(10, 1/2)$,

$$\begin{aligned}P(X = 8) &= \binom{10}{8} \left(\frac{1}{2}\right)^8 \left(1 - \frac{1}{2}\right)^{10-8} \\&= \frac{10!}{8! 2!} \left(\frac{1}{2}\right)^{10} \\&= \frac{45}{1024} \\&\approx 0.0439453.\end{aligned}$$

Let's verify the calculation in RStudio.

- First compute it from factorials.
- Then use R's built-in binomial pmf.

Probability 21

Computing the binomial pmf manually in R

```
n <- 10
k <- 8
p <- 0.5

n_factorial <- factorial(n)
k_factorial <- factorial(k)
nmk_factorial <- factorial(n - k)

(n_factorial / (k_factorial * nmk_factorial)) *
  p^k * (1 - p)^(n - k)

[1] 0.04394531
```


Probability 22

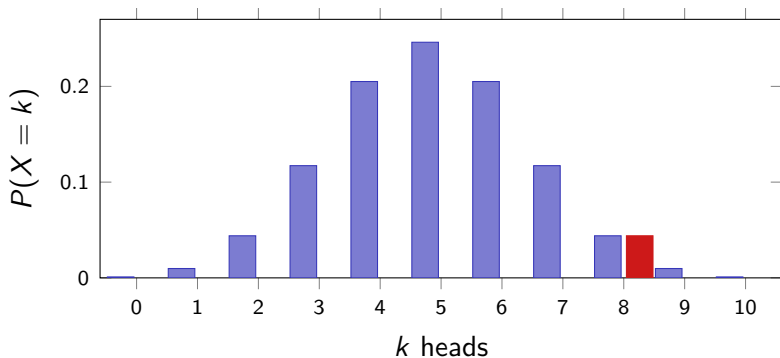
The built-in R function

```
> dbinom(8, size = 10, prob = 0.5)
[1] 0.04394531
```

In R, `dbinom()` evaluates the binomial *pmf* at a specified count.

Probability 23

Where eight heads sits in the distribution



$$P(X = 8) = 0.0439453$$

What is the probability of eight or fewer heads in ten fair tosses?

Binomial CDF

If $X \sim \text{Binomial}(n, p)$, then

$$F_X(k) = P(X \leq k) = \sum_{i=0}^k \binom{n}{i} p^i (1-p)^{n-i}.$$

For $n = 10$, $p = 1/2$, and $k = 8$, this adds the pmf values from 0 through 8.

```
> pbinom(8, size = 10, prob = 0.5)
[1] 0.9892578
```

$$P(X \leq 8) = 0.9892578.$$

In R, `pbinom()` evaluates the binomial *CDF*; `dbinom()` evaluates the pmf.

Probability 26

Continuous random variables

- A continuous random variable has a CDF F_X that is continuous.
- For every fixed x , $P(X = x) = 0$; probabilities are assigned to intervals.
- If X has density f_X , then

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1,$$

and for $a \leq b$,

$$P(a < X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x) dx.$$

Let X be the waiting time, in seconds, between incoming calls. Suppose

$$f_X(x) = \begin{cases} \frac{1}{4}e^{-x/4}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

This is a valid density because $f_X(x) \geq 0$ and

$$\int_0^{\infty} \frac{1}{4}e^{-x/4} dx = 1.$$

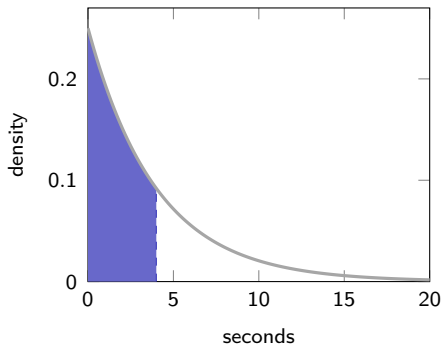
The tail probability beyond four seconds is

$$P(X > 4) = \int_4^{\infty} \frac{1}{4}e^{-x/4} dx = e^{-1} \approx 0.3679.$$

Probability 28

What about a wait between 0 and 4 seconds?

$$\begin{aligned}P(0 \leq X \leq 4) &= \int_0^4 \frac{1}{4} e^{-x/4} dx \\&= 1 - e^{-1} \\&\approx 0.6321.\end{aligned}$$



A normal random variable with mean μ and variance σ^2 has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right], \quad \sigma > 0.$$

We write

$$X \sim \mathcal{N}(\mu, \sigma^2).$$

The parameter μ locates the center; σ is the standard deviation and controls the spread.

Probability 30

The LLN says the sample mean settles near the population mean

Let $X_1, \dots, X_n$ be independent and identically distributed with $\mathbb{E}[X_i] = \mu$. The sample mean is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Law of Large Numbers

As the sample size grows,

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty.$$

- The LLN explains why averages and relative frequencies stabilize.
- It concerns where $\bar{X}_n$ goes; it does not specify the shape of its sampling distribution.

Probability 31

The CLT describes the distribution of sample means

Suppose additionally that $\text{Var}(X_i) = \sigma^2 < \infty$. Across hypothetical repeated samples of size n ,

Central Limit Theorem

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

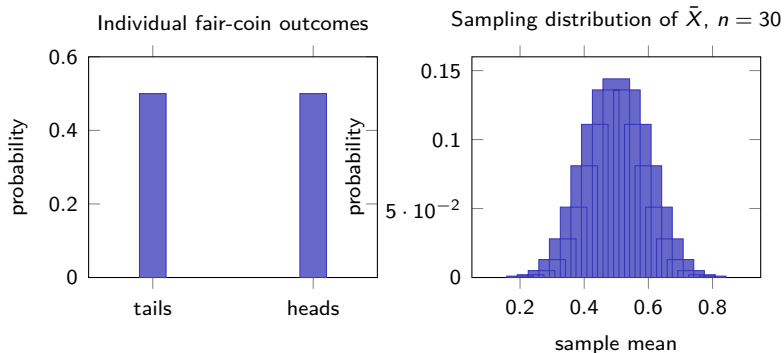
For sufficiently large n , the sampling distribution is approximately

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

- Center: μ . Standard error: $\sigma/\sqrt{n}$.
- The CLT does not say that the individual observations become normally distributed.

Probability 32

A Bernoulli population can yield an approximately normal mean



Political science translation: A sample turnout rate or support proportion is the mean of Bernoulli observations, so the same sampling-distribution logic applies.

The quality-control problem

The machine made a suspicious *thud*. I rolled one die 30 times, obtained a sample mean of 3.30, and declared that there was “no difference.”

Were 30 rolls enough to detect a small defect?

Probability 34

A defect that shifts the average roll

Suppose the true pmf is

$$p_X(x) = \begin{cases} 0.20, & x = 1, \\ 0.20, & x = 2, \\ 0.20, & x = 3, \\ 0.15, & x = 4, \\ 0.10, & x = 5, \\ 0.15, & x = 6, \\ 0, & \text{otherwise.} \end{cases} \quad \sum_{x=1}^6 p_X(x) = 1.$$

Its mean is $\mathbb{E}[X] = 3.2$, not the fair-die mean 3.5.

For this example, we focus on whether sample data can detect this mean shift. That is narrower than testing whether all six faces have probability $1/6$.

Probability 35

A sample helps us learn about an unknown population

We usually cannot observe every member of a population. Instead, we study a sample and use it to learn about the population.

Political science question: does campaign contact increase turnout?

Population question: Among all voters of interest, how different would turnout be under campaign contact versus no contact?

Sample evidence: In the study sample, compare turnout in the contact and no-contact groups.

From sample to population: Results would change if we repeated the study with a new sample. Statistical inference uses this expected sampling variation to describe how precisely the sample difference estimates the population difference.

Design matters: random assignment supports a causal interpretation; otherwise the comparison describes an association.

Probability 36

The sample mean targets one feature of the distribution

In the die example, the unknown population mean μ is the long-run average roll. The sample mean $\bar{X}$ is our estimate of μ . A fair die gives the benchmark

$$\mu_0 = \mathbb{E}[X] = 3.5, \quad \sigma_0 = \sqrt{\text{Var}(X)} = \sqrt{\frac{35}{12}} \approx 1.7078.$$

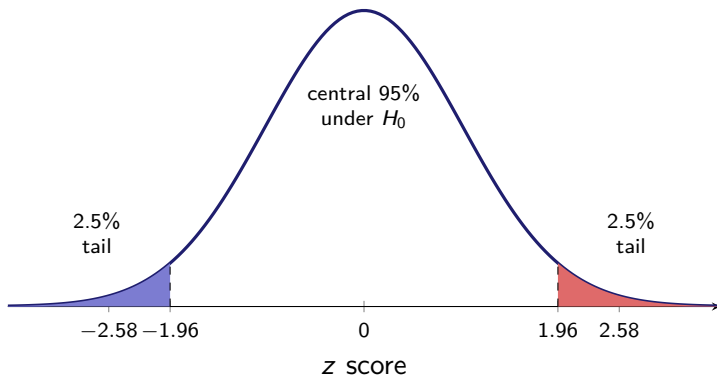
Across repeated samples of n rolls, $\bar{X}$ varies. Its standard error describes that sampling variation, and Z measures the sample mean's distance from 3.5 in standard-error units:

$$\text{SE}_0(\bar{X}) = \frac{\sigma_0}{\sqrt{n}}, \quad Z = \frac{\bar{X} - \mu_0}{\text{SE}_0(\bar{X})}.$$

If $\bar{X}$ is many standard errors from 3.5, the sample suggests that the population mean is not 3.5. This test addresses a mean shift, not whether every face has probability $1/6$.

Probability 37

Z scores turn unusual sample means into tail areas



Probability 38

Thirty rolls were not enough to detect the mean shift

Suppose my 30 rolls produced $\bar{X} = 3.30$. Under the benchmark hypothesis $H_0 : \mu = 3.5$,

$$SE_0(\bar{X}) = \frac{\sqrt{35/12}}{\sqrt{30}} \approx 0.312,$$

$$Z = \frac{3.30 - 3.50}{0.312} \approx -0.64, \quad p\text{-value} \approx 0.52.$$

Decision and interpretation

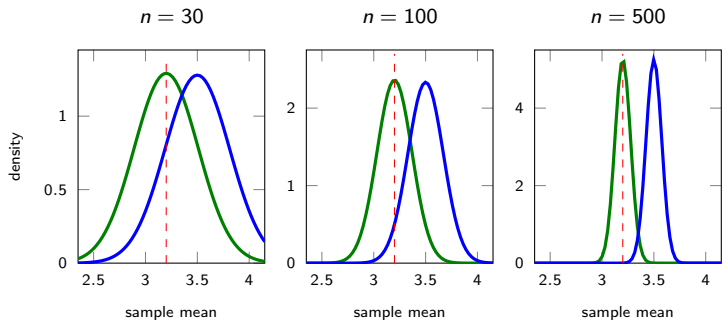
We **fail to reject** $H_0 : \mu = 3.5$. These 30 rolls do not provide strong evidence that the defect shifted the mean.

We have not shown that the mean is 3.5—or that the die is fair.

My 14-year-old self called the machine fixed and went back to sleep.

Probability 39

Larger samples separate competing mean models



defect model: $\mu = 3.2$

benchmark model: $\mu = 3.5$

true defect mean

Probability 40

The mean shift becomes detectable with enough rolls

Suppose the larger samples produce means close to the defect-model mean of 3.20.

n	$\bar{X}$	SE_0	Z	p	Decision at $\alpha = .05$
30	3.30	.312	-.64	.521	Fail to reject
100	3.20	.171	-1.76	.079	Fail to reject
500	3.20	.076	-3.93	< .001	Reject

What changes as n grows?

The hypothesized mean difference remains 0.30, but the standard error shrinks. At $n = 100$, the evidence still falls short of the conventional threshold; at $n = 500$, the same mean shift is clearly distinguishable from 3.5.

Probability 41

Changing the question changes what we test

Did the average roll change?

$$H_0 : \mu = 3.5$$

Statistic: the sample mean $\bar{X}$.

Target: a **mean shift**.

Did any face probability change?

$$H_0 : p_1 = \cdots = p_6 = \frac{1}{6}$$

Statistic: all six face counts.

Target: **any departure** from the fair-die pmf.

Different questions, different evidence

A die can have mean 3.5 without being fair. The full-distribution question is therefore broader than the mean question.

Probability 42

A chi-square test uses all six face counts

Let O_j be the observed number of rolls showing face j . Under a fair die,

$$E_j = \frac{n}{6}, \quad \chi^2 = \sum_{j=1}^6 \frac{(O_j - E_j)^2}{E_j}.$$

Under H_0 , the statistic is approximately

$$\chi^2 \sim \chi_5^2,$$

because six face probabilities are compared subject to the requirement that they sum to one.

With only 30 rolls

Each expected count is $E_j = 5$, which is just at the usual rule-of-thumb boundary for the chi-square approximation. A simulated p -value avoids leaning heavily on that approximation.

```
In R: chisq.test(counts, p = rep(1/6, 6), simulate.p.value = TRUE, B = 10000)
```

Probability 43

A different defect can leave the mean unchanged

For the full-distribution question, consider a second hypothetical defect:

$$\mathbf{p}_B = (0.25, 0.15, 0.10, 0.10, 0.15, 0.25).$$

Face	1	2	3	4	5	6
Fair-die probability	.167	.167	.167	.167	.167	.167
Defect-model probability	.25	.15	.10	.10	.15	.25

Because the probabilities are symmetric around 3.5,

$$\mathbb{E}_B[X] = 3.5.$$

Why change the question?

A mean test cannot detect this defect because its population mean is unchanged. A chi-square test can detect the redistribution across faces.

Probability 44

Thirty rolls still do not clearly reveal the defect

Suppose the defect produces 30 rolls with counts close to its expected pattern:

Face	1	2	3	4	5	6
Observed count O_j	8	4	3	3	4	8
Fair-die expected count E_j	5	5	5	5	5	5

$$\bar{X} = 3.5, \quad X^2 = 5.60, \quad p_{\chi^2} \approx .347, \quad p_{\text{sim}} \approx .368.$$

Decision and interpretation

At $\alpha = .05$, we **fail to reject** uniform face probabilities. The sample does not clearly distinguish the defect from a fair die, but it does not establish that the die is fair.

Probability 45

With 100 rolls, the distributional pattern becomes clearer

Now suppose 100 rolls reproduce the same defect pattern more precisely:

Face	1	2	3	4	5	6
Observed count O_j	25	15	10	10	15	25
Fair-die expected count E_j	16.67	16.67	16.67	16.67	16.67	16.67

$$\bar{X} = 3.5, \quad \chi^2 = 14.00, \quad p_{\chi^2} \approx .016, \quad p_{\text{sim}} \approx .016.$$

Decision and interpretation

At $\alpha = .05$, we **reject** uniform face probabilities. The larger sample provides enough information to distinguish this nonuniform pmf from the fair-die benchmark.

Probability 46

With 500 rolls, the evidence becomes decisive

Finally, suppose 500 rolls reproduce the same nonuniform pmf:

Face	1	2	3	4	5	6
Observed count O_j	125	75	50	50	75	125
Fair-die expected count E_j	83.33	83.33	83.33	83.33	83.33	83.33

$$\bar{X} = 3.5, \quad \chi^2 = 70.00, \quad p < .001.$$

Decision and interpretation

We again **reject** uniform face probabilities, now with much stronger evidence. The larger sample estimates the underlying face probabilities more precisely, while the mean remains 3.5.

Probability 47

What my 14-year-old self should have learned

- Samples vary even when the underlying process does not.
- Mean and chi-square tests can reach different conclusions because they test different hypotheses.
- Failing to reject H_0 does not establish that the benchmark is correct.
- Larger samples reduce uncertainty and make small departures easier to detect.

The central lesson

Evidence against a benchmark depends on both the question being asked and how much information the sample contains.

A nonsignificant result is not permission to go back to sleep.